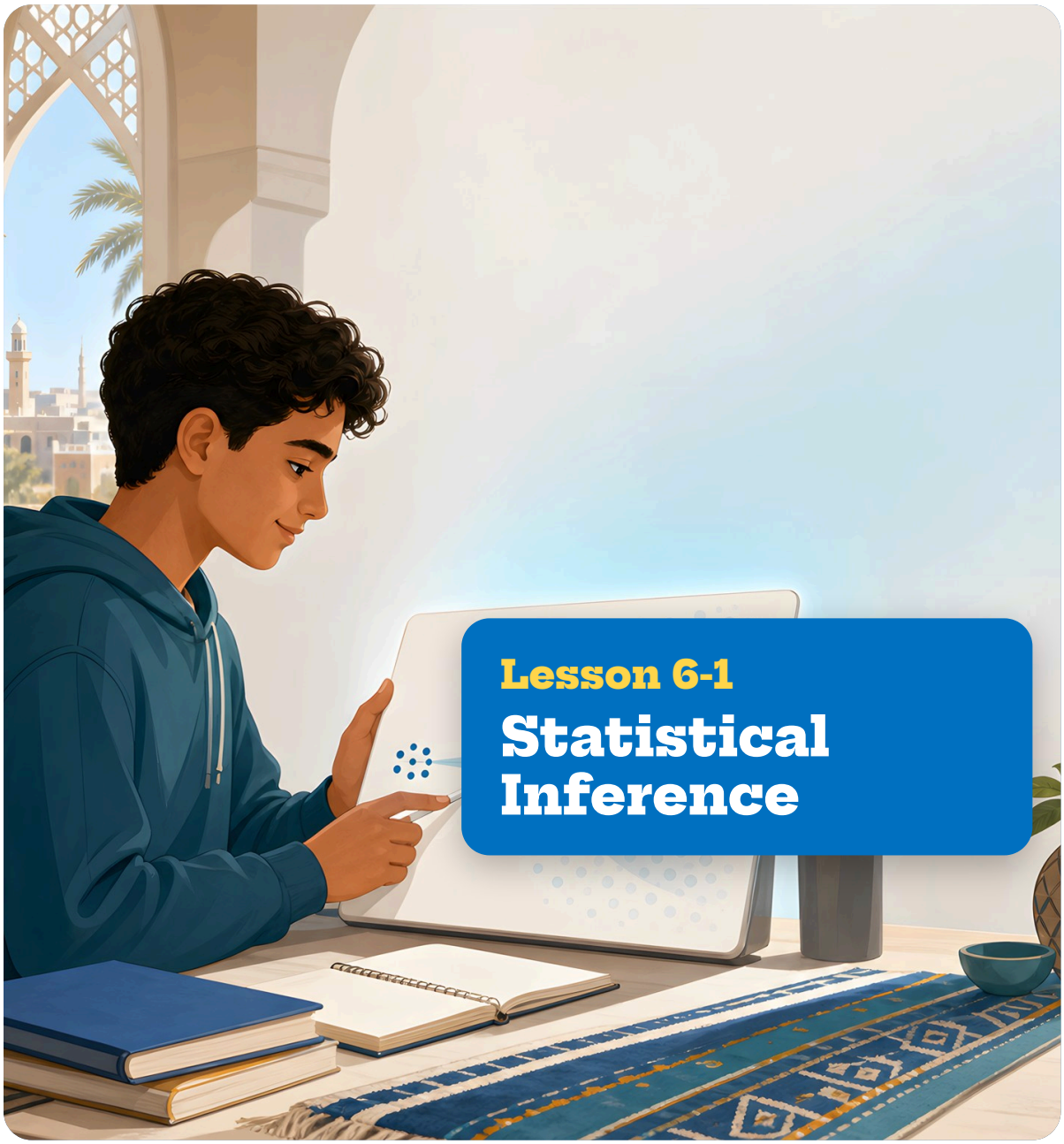


Learning Objectives

- Explain the difference between point estimation and interval estimation, including the meaning of a confidence level.
- Describe the procedure of hypothesis testing, and explain the cautions needed when interpreting its results.

You can never survey everyone, so you study a sample and reason back to the whole population — that is statistical inference. It answers two kinds of question: "about how big?" (estimation) and "is this real, or just chance?" (hypothesis testing). But because a sample is only part of the whole, every answer carries uncertainty. The skill is not to hide that uncertainty but to measure it — with confidence intervals, p-values, and significance levels — and to remember that "statistically significant" is not the same as "important."

? Guiding question: How can we reason from a sample to a whole population, and how do we measure the uncertainty in that reasoning?

📌 Point!

1. The Statistical Inference Stage

(1) A sample is drawn from the entire group that is the target of analysis (the population) and surveyed, and the properties of the population are then inferred from the results of the sample. Inferring the properties of the population from the sample in this way is called **(statistical inference)**.

(2) Statistical inference can be broadly divided into the following two approaches.

Approach	Description	Example
(Estimation)	Infer the properties of the population in numerical terms	Infer the population mean height from the sample mean height
(Hypothesis testing)	Determine whether a hypothesis about the population is correct	Determine whether a new learning method is more effective than the conventional one

(3) In either case, since the sample is only part of the population, inference always involves uncertainty. In statistical inference, judgment is made while considering the degree of that uncertainty.

2. Point Estimation and Interval Estimation

(1) **(Point estimation)**…estimating the properties of the population as a single value calculated from the sample. *e.g.,* If the sample mean height is 168 cm, then the population mean height is also estimated to be 168 cm.

(2) **(Interval estimation)**…estimating that the properties of the population fall within a certain range (interval). *e.g.,* The population mean height is estimated to be in the range of 166 cm to 170 cm.

(3) The interval used in interval estimation is called the **(confidence interval)**, and how often such intervals contain the population value over repeated surveys is called the **(confidence level)**. Levels of 95% or 99% are common.

Key Fact:
Statistical inference reasons from a sample to a population. Estimation gives a value (point) or a range (interval, with a confidence level); hypothesis testing weighs a null against an alternative using a p-value and a significance level. All of it carries uncertainty — and significance is not the same as practical importance.

Key Concepts:

- statistical inference
- point / interval estimation
- confidence interval & level
- null / alternative hypothesis
- p-value & significance level
- Type I / II errors

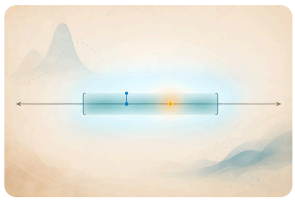
KEY TERMS
statistical inference — reasoning from a sample to the population.
estimation — inferring the population in numbers.
hypothesis testing — judging a claim about the population.

💡 PAUSE & THINK

A single value or a range — which estimate shows you how uncertain it is?

e.g., Suppose the population mean height is estimated by sample survey, and a confidence interval of “166 cm to 170 cm” is obtained at a 95% confidence level. The “95%” means that if the same survey method is repeated 100 times, about 95 of the resulting confidence intervals will contain the true population mean height.

(4) Point estimation is easy to understand because it is shown as a single value, but the uncertainty of the estimate is invisible. Interval estimation has width, so care is needed in interpreting the result, but it has the advantage of also showing the degree of uncertainty.



A point estimate is one value; an interval is a range.

3. The Idea of Hypothesis Testing

(1) Hypothesis testing…a procedure for determining whether a hypothesis about the population is acceptable based on the data at hand.

(2) In hypothesis testing, the following two hypotheses are set up.

Hypothesis	Description
(Null hypothesis) (H0)	A hypothesis serving as the starting point of the test, such as “no difference” or “no relationship”
(Alternative hypothesis) (H1)	The hypothesis adopted when the null hypothesis is rejected, such as “there is a difference” or “there is a relationship”

(3) In hypothesis testing, the probability that data collection could result in the data at hand is evaluated under the assumption that “there is no difference or relationship.” The reason for first assuming the hypothesis one wants to deny (the null hypothesis) is that, rather than directly showing “there is a difference or relationship,” it is logically easier to judge this by showing that, under the assumption “there is no difference or relationship,” the data at hand would be unlikely to be obtained. *e.g.,* It is difficult to directly show “the new learning method is effective.” Instead, one assumes “there is no effect” and then examines how low the probability is of obtaining a difference like the experimental result at hand, and if that probability is sufficiently small, one judges that “no effect” is unlikely.

(4) Based on the evaluation, the null hypothesis is judged to be either **(rejected (denied))** or **(not rejected)**.



Testing weighs evidence against the null hypothesis.

KEY TERMS
null hypothesis (H0) — the “no difference” starting point.
alternative hypothesis (H1) — adopted if H0 is rejected.
p-value — chance of data at least this extreme if H0 holds.

4. The Procedure of Hypothesis Testing

(1) Hypothesis testing follows this flow.

① Set the null hypothesis H0 and the alternative hypothesis H1

② Decide the **(significance level)** (5% = 0.05 and 1% = 0.01 are commonly used)

③ Compute the **(p-value)** from the data at hand

④ Compare the p-value with the significance level and make a judgment

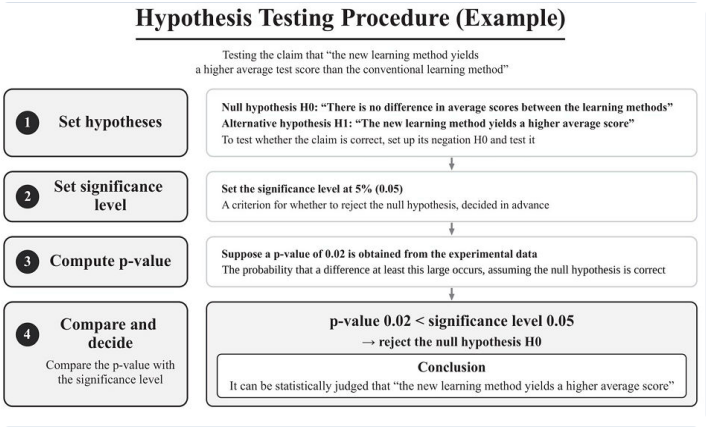
(2) p-value…assuming the null hypothesis is correct, the probability of obtaining a result as extreme as, or more extreme than, the data at hand. The

smaller the p-value, the less likely it is that the data would be obtained under the null hypothesis.

(3) Significance level…the threshold value used to decide whether to reject the null hypothesis.

(4) Criteria for judgment

Condition	Judgment
p-value < significance level	Reject the null hypothesis and adopt the alternative hypothesis (statistically significant)
p-value ≥ significance level	Do not reject the null hypothesis (cannot be said to be statistically significant)



The hypothesis testing procedure worked through an example

5. Major Hypothesis Testing Methods

(1) The test method used differs depending on the purpose of the analysis and the type of data. Major examples include the following.

Test	Main use	Example
(t-test)	Examine whether there is a difference in the means of two groups	Examine whether students' average test scores differ between a certain learning method and the conventional method
(Chi-squared test)	Examine whether there is a relationship between two categorical variables (data expressed by classification, such as gender, occupation, or preference)	Examine, from survey results, whether there is a relationship between gender and subject preference

6. Cautions Regarding Inference

(1) In statistical inference, there is always a possibility of judgment errors. There are two types of errors.

Type of error	Description
(Type I error)	Rejecting the null hypothesis when it is actually correct
(Type II error)	Not rejecting the null hypothesis when it is actually incorrect

PAUSE & THINK

If the p-value is 0.02 and the significance level is 0.05, do you reject the null or not?

PAUSE & THINK

Which test compares two group means, and which checks a relationship between categories?

KEY TERMS

Type I error — rejecting a true null hypothesis.
Type II error — keeping a false null hypothesis.
trade-off — lowering one error raises the other.

(2) Lowering the significance level reduces the Type I error, but tends to increase the Type II error. The two have a **(trade-off relationship)**.

(3) Care is needed in interpreting the p-value. The following misunderstandings must be avoided.

① The p-value is not “the probability that the null hypothesis is correct.”

② A small p-value does not mean “the effect is large.”

③ The comparison of the p-value with the significance level is not an absolute correctness criterion but a guide for judgment. The significance level (e.g., 0.05 or 0.01) is a value chosen according to the purpose of the analysis or the conventions of the field; falling below this value does not mean the result is certainly correct.

(4) Being statistically significant and being practically important are different. *e.g.,* Suppose an experiment compares two designs of a website, with click-rate data from 100,000 people split evenly. The old design is clicked by 0.4% of visitors, the new one by 0.5% — only 0.1 percentage points higher. Because the data set is large, this difference is likely to be statistically significant. However, whether that 0.1-point difference leads to actual profit or better user experience needs to be judged from a separate perspective.

(5) Statistical inference is a powerful means of drawing conclusions from data, but it is important to understand that it is a tool for judgment that includes uncertainty, and the results should be interpreted appropriately.



Type I is a false alarm; Type II is a miss.

PAUSE & THINK

With 100,000 people, a tiny difference can be “significant.” Does that make it important?

Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Inferring the properties of the population from the sample is called statistical inference.

B Point estimation is a method that estimates the properties of the population within a certain range.

C In hypothesis testing, the probability that data collection could result in the data at hand is evaluated under the assumption that the null hypothesis is correct.

D If a result is statistically significant, the difference is also necessarily important in practice.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A hypothesis serving as the starting point of the test, such as “no difference” or “no relationship”

b The probability of obtaining a result as extreme as, or more extreme than, the data at hand, assuming the null hypothesis is correct

c The threshold value for deciding whether to reject the null hypothesis

[Choices] **A** p-value **B** Significance level **C** Null hypothesis

✓ Solution

(1)

A Correct definition of statistical inference. Therefore, ○.

B Estimating within a range is interval estimation. Point estimation estimates a single value. Therefore, ×.

C Correct description of the idea of hypothesis testing. Therefore, ○.

D Statistical significance and practical importance are different; with a large amount of data, even a tiny difference can become significant. Therefore, ×.

(2)

a: A hypothesis serving as the starting point of the test is the null hypothesis. **C**

b: The probability of obtaining the data under the null hypothesis is the p-value. **A**

c: The threshold value for rejection is the significance level. **B**

Try

1 Answer the following questions.

1. What is the term for inferring the properties of the population from the sample?
2. What is the term for the method of estimating that the properties of the population fall within a certain range?
3. What is the term for the interval used in interval estimation?
4. What is the term for a typical hypothesis testing method that examines whether there is a relationship between two categorical variables?

EXAM-STYLE QUESTION

Explain why researchers test a null hypothesis they hope to disprove, rather than proving the effect directly. **[6] Refer to:** what a small p-value shows.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the procedure of hypothesis testing.

A Set the null and alternative hypotheses → decide the significance level → compute the p-value → make a judgment

B Compute the p-value → set the null hypothesis → decide the significance level → make a judgment

C Decide the significance level → adopt the alternative hypothesis → compute the p-value → set the null hypothesis

D Make a judgment → set the null hypothesis → decide the significance level → compute the p-value

(2) Suppose that, in a hypothesis test with the significance level set at 0.05 (5%), the p-value is 0.03. From the following options (A–D), choose the one that most appropriately describes the judgment in this case.

A Reject the null hypothesis and adopt the alternative hypothesis.

B Do not reject the null hypothesis.

C Reject the alternative hypothesis.

D Change the significance level to 0.01 and recalculate.

(3) For each of analysis purposes (a, b), choose the most appropriate test method from the choices below (A, B).

a You want to examine whether students' average test scores differ between two teaching methods.

b You want to examine whether there is a relationship between gender and the category of products purchased.

[Choices] **A** Chi-squared test **B** t-test

(4) From the following options (A–D), choose the one that most appropriately describes the p-value.

A The p-value is the probability that the null hypothesis is correct.

B The smaller the p-value, the larger the effect is.

C The p-value is the probability of obtaining a result as extreme as, or more extreme than, the data at hand, assuming the null hypothesis is correct.

D p-value < 0.05 is an absolute criterion of correctness.

Think as an Engineer — Investigate & Decide

A school club claims its new online quiz tool raises students' average score on weekly tests. You are asked to judge the claim fairly using a hypothesis test.

☐ **Trust the club's own summary** ☐ **Frame and test the claim properly**

PAUSE & THINK

p-value 0.03 against a 0.05 threshold — below or above? So what do you do?

KEY TERMS

significance level — the threshold for rejecting H0.
statistically significant — p below the threshold.
practically important — big enough to matter in real life.

1 · Frame the hypotheses. The club reports its tool group averaged 74 on the last quiz versus 70 for a control group of the same size. Write the null and alternative hypotheses for this exact claim in your own words, state what data you would still need to actually run the test (beyond these two means), and say which significance level you would set (0.05 or 0.01) and why.

2 · Analyze the risks. Describe, in this context, what a Type I error and a Type II error would each mean for the club, and say which one you would most want to avoid here — a judgment the lesson's tables do not make for you.

3 · Decide and interpret. Suppose the test returns a p-value of 0.04. State your decision against the significance level you chose in step 1 (note it would differ at 0.05 versus 0.01). Then, if the result is significant, explain why even a significant result might still not justify switching every class to the tool. **In pairs,** compare interpretations and agree on what evidence beyond the p-value you would still want.

Give evidence for your answer.

Stuck? Start from what you are trying to disprove: the null is always “no difference.” Then ask what it would cost to reject it wrongly (Type I) versus to miss a real effect (Type II) — that cost is what should guide your significance level.

🌍 In a New Context

A shop tests two checkout-button colors with data from 200,000 visitors and finds the new color's click rate is 0.05 percentage points higher, with a p-value of 0.01 at the 5% significance level. State whether the result is statistically significant, and explain why the shop might still decide not to change the button.

❓ Lesson Question — Answered

The lesson asked how we reason from a sample to a whole population and how we measure the uncertainty in that reasoning. Because a sample is only part of the population, we use statistical inference — either estimation, which reports the population as a single value (point estimation) or as a range (interval estimation) whose confidence level says how often such intervals would contain the true value, or hypothesis testing, which sets a null hypothesis ("no difference") against an alternative and compares a p-value with a chosen significance level to decide whether to reject the null. The uncertainty is never removed: it is measured and reported. Judgment can fail as a Type I error (rejecting a true null) or a Type II error (keeping a false null), and lowering one raises the other. And a result that is statistically significant — especially with very large samples — is not automatically important in practice, so results must be interpreted, not just computed.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

In statistical inference, the properties of the population are inferred from the sample. Estimating the properties of the population as a single value is called (a), and estimating within a certain range is called (b). Also, the procedure of judging a hypothesis about the population is called (c). In hypothesis testing, the (d)—such as "no difference" or "no relationship"—and the opposite (e) are set up, and a judgment is made by comparing the (f) computed from the data at hand with the predetermined (g).

1. Fill in the blanks (a)–(g).

⭐ REFLECT & CHALLENGE

Reflect: Why is a 95% confidence level a statement about the method, not about one particular interval?
Challenge: A study of 500,000 users reports a "highly significant" effect ($p < 0.001$) whose size is 0.02 percentage points. Argue whether you would act on it, and name the one extra piece of information that would change your mind.

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes confidence intervals and confidence levels.

A A confidence interval with a 95% confidence level means that the true population value is contained in that interval with a probability of 95%.

B The higher the confidence level, the narrower the confidence interval.

C A confidence interval with a 95% confidence level means that, if the same survey method is repeated, about 95% of the resulting confidence intervals will contain the true population value.

D The confidence level is a value used as the criterion for judgment in hypothesis testing.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of errors in hypothesis testing.

A A Type I error is rejecting the null hypothesis when it is actually correct.

B A Type II error is not rejecting the null hypothesis when it is actually incorrect.

C Lowering the significance level can simultaneously reduce both Type I and Type II errors.

D Type I and Type II errors have a trade-off relationship.

(3) From the following options (A–D), choose the one that most appropriately describes the interpretation of statistical inference.

A A statistically significant difference is necessarily an important difference in practice.

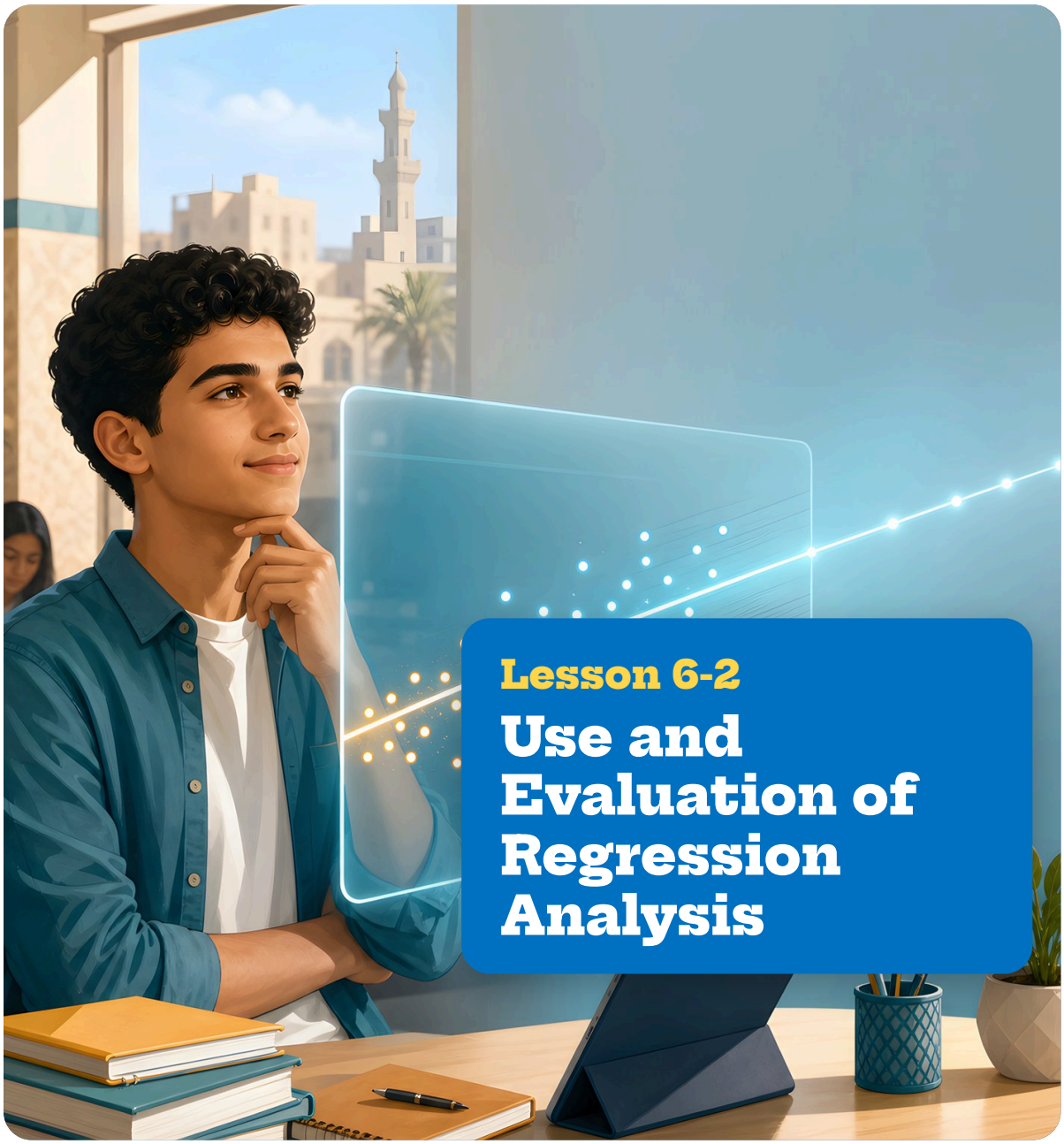
B With a large amount of data, even a tiny difference can become statistically significant.

C If the p-value falls below 0.05, the conclusion is certainly correct.

D Statistical inference can completely remove uncertainty.

⭐ Key takeaway

Remember:
Estimate with a point or an interval (confidence level = how often the method captures the truth); test by comparing a p-value with a significance level to reject or keep the null. Watch Type I / II errors, and never confuse statistical significance with practical importance.



Learning Objectives

- Explain the difference between simple and multiple regression, and what a regression model represents.
- Evaluate a regression model using R^2, RMSE, and MAE, and explain why a good fit does not prove causation.

Data often hides a relationship you can turn into a prediction: the warmer the day, the more ice cream sells. Regression analysis writes that relationship as an equation, so you can put in tomorrow's forecast temperature and predict tomorrow's sales. Yet a prediction equation is only a model — a simplified picture — and a careful analyst asks two more questions: how accurate is it (using R^2, RMSE, and MAE), and does "fits well" really mean "causes"? A line that fits the data is not proof of cause.

? Guiding question: How does regression turn data into a prediction, and how do we judge whether that prediction can be trusted?

Point!

1. The Regression Analysis Stage

(1) **(Regression analysis)**…a method that expresses the relationship between variables in the data as an equation, and uses it for prediction and simulation. For example, from past data on temperature and sales, one can express how sales change when temperature rises as an equation, and put tomorrow's forecast temperature into that equation to predict sales.

(2) In regression analysis, the variable that is the result is called the **(response variable) (dependent variable)**, and the variable used to explain or predict it is called the **(explanatory variable)**. In the temperature-and-sales example, sales are the response variable and temperature is the explanatory variable.

(3) The equation obtained from data in regression analysis is called the **(model)**. A model is a simplified representation of a real relationship—for example, putting the trend "the higher the temperature, the higher the sales" into a format that can be calculated.

(4) Both creating a model and evaluating whether the model can actually be used are important. A model that is only created is not useful; only by confirming its accuracy and limitations does it lead to practical use.

2. Simple and Multiple Regression

(1) **(Simple regression analysis)**…a method of predicting the response variable using one explanatory variable.

e.g., Predict ice cream sales (y) from temperature (x) alone. The prediction equation is expressed as a straight line in the format **$y = ax + b$**, and this line is called the **(regression line)**. a and b are determined by the **(least-squares method)** (a method that determines the line so as to minimize the sum of the squares of the discrepancies between the actual values and the regression line, each called a **(residual)**).

(2) **(Multiple regression analysis)**…a method of predicting the response variable using multiple explanatory variables. In real data, results are rarely determined by a single cause, and multiple regression analysis, which can consider several factors simultaneously, is suited to more realistic prediction.

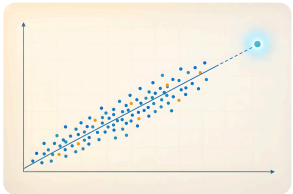
Key Fact:
Regression analysis expresses the relationship between an explanatory variable (the input used to make the prediction) and a response variable (the result) as a model — a line $y = ax + b$ for simple regression, or a sum of weighted variables for multiple regression. Judge the model with R^2, RMSE, and MAE, and never mistake a good fit for proof of causation.

Key Concepts:

- regression analysis
- response / explanatory variable
- simple / multiple regression
- regression line, least-squares
- R^2, RMSE, MAE
- correlation versus causation

KEY TERMS

regression analysis — expressing a relationship as an equation for prediction.
response variable — the result (dependent).
explanatory variable — the input used to make the prediction.



Fit a line to data, then use it to predict within that range.

e.g., Predict housing prices from a house's size, age, and distance from a transit station (multiple factors). The prediction equation is expressed as the sum of multiple explanatory variables, each multiplied by a coefficient (a numerical value representing the strength of its influence), plus an intercept. For example, when there are two explanatory variables, the format becomes $\boldsymbol{y = a_1x_1 + a_2x_2 + b}$ (x_1 and x_2 are the explanatory variables, a_1 and a_2 are their coefficients, and b is the intercept). When there are three or more explanatory variables, terms are added in the same way.

(3) The following differences exist between simple and multiple regression.

Aspect	Simple regression	Multiple regression
Number of explanatory variables	One	Two or more
Characteristics of the model	Simple and easy to interpret	Can express complex relationships
Suitable situations	When you want to capture the relationship between two variables concisely	When the response variable is influenced by multiple factors

3. Evaluation Indicators for Models

(1) To judge how useful a model created by regression analysis is, **(evaluation indicators)** are used. Three typical indicators are as follows.

Indicator	Description (concept of calculation)	Interpretation
(R^2 (coefficient of determination))	Expresses, on a 0–1 scale, how well the model explains the variability in the data	The closer to 1, the better the variation in the data is explained
(RMSE (root mean square error))	The square root of the mean of the squared discrepancies between predicted and actual values	The smaller the value, the higher the prediction accuracy; expressed in the same units as the response variable
(MAE (mean absolute error))	The mean of the absolute values of the discrepancies between predicted and actual values	The smaller the value, the higher the prediction accuracy; less affected by extremely large discrepancies

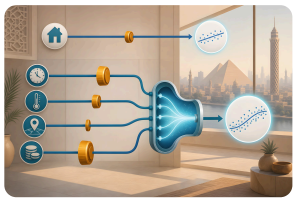
(2) Both RMSE and MAE indicate the size of prediction errors, but RMSE tends to weight large discrepancies more heavily. On the other hand, MAE evaluates discrepancies on average, so it is less affected by outliers. They are used differently according to the purpose of the analysis.

4. Cautions When Using Regression Models

(1) Regression analysis is a widely applicable method, but misuse can lead to incorrect conclusions. Pay particular attention to the following points.

PAUSE & THINK

One cause or several? Which kind of regression would house prices need?



Multiple regression combines several inputs.

KEY TERMS

R^2 — how well the model explains the variability in the data (0–1).
RMSE — error size, weights big misses more.
MAE — average error, less swayed by outliers.

PAUSE & THINK

A prediction line fits the data well — does that prove one variable causes the other?

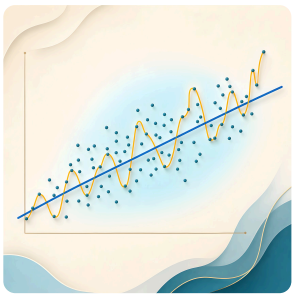
① Do not confuse correlation with causation: a prediction equation that fits does not establish it. Causation needs temporal precedence, exclusion of other factors' influence, and theoretical support.

② A model is not finished once created: Confirm accuracy with evaluation indicators, verify validity in light of the purpose, and do not apply it outside the range of the data it was fitted to.

③ Do not judge a model by R^2 alone: Make judgments together with other indicators.

④ Adding more explanatory variables does not necessarily improve a model: Appropriate variables must be chosen according to the purpose.

(2) The results of regression analysis should not be treated as absolute truth as is, but as one perspective that the data offers. Interpreting results appropriately and using them in real-world judgment is important.



An overfit model chases noise, not the trend.

PAUSE & THINK

Which error measure would you trust more when a few huge outliers exist?

Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Multiple regression analysis is a method of predicting the response variable using one explanatory variable.

B R^2 (coefficient of determination) is an indicator showing how well the model explains the variability in the data.

C The equation obtained from data in regression analysis is called the model, and is a simplified representation of a real relationship.

D If the prediction equation fits, it can be said that there is a causal relationship between the explanatory variable and the response variable.

(2) Match each description (a–c) with the most appropriate indicator from the choices below (A–C).

a The mean of the absolute values of the discrepancies between predicted and actual values

b The square root of the mean of the squared discrepancies between predicted and actual values

c How well the model explains the variability in the data

[Choices] **A** RMSE **B** R^2 **C** MAE

Solution

(1)

A Predicting with one explanatory variable is simple regression analysis. Multiple regression analysis uses multiple explanatory variables. Therefore, ×.

B Correct description of R^2. Therefore, ○.

C Correct definition of a model. Therefore, ○.

D Even if the prediction equation fits, that alone does not establish causation. Therefore, ×.

(2)

a: The mean of absolute discrepancies is MAE. **C**

b: The square root of the mean of squared discrepancies is RMSE. **A**

c: An indicator that explains the variability in the data is R^2. **B**

Try

1 Answer the following questions.

1. What is the term for the regression analysis that predicts the response variable using multiple explanatory variables?
2. What is the term for the indicator showing how well the model explains the variability in the data?
3. In regression analysis, what are the variables called that are the result and the input used to predict it, respectively?
4. What is the term for "the discrepancy between actual values and the regression line" used in simple regression analysis?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the difference between simple and multiple regression.

A Simple regression can express complex relationships, while multiple regression is simple.

B Simple regression uses multiple explanatory variables, and multiple regression uses one explanatory variable.

C Simple regression uses one explanatory variable, and multiple regression uses multiple explanatory variables.

D There is no difference between simple and multiple regression in the number of explanatory variables; only the calculation method differs.

(2) For each of analysis purposes (a, b), choose the most appropriate regression analysis method from the choices below (A, B).

a You want to predict ice cream sales from temperature (one variable).

b You want to predict housing prices from a house's size, age, and distance from a transit station (multiple variables).

[Choices] **A** Simple regression analysis **B** Multiple regression analysis

EXAM-STYLE QUESTION

Explain why a regression model with a high R^2 can still be a poor basis for claiming that one variable causes another. **[6] Refer to:** correlation versus causation.

PAUSE & THINK

One variable in, or several? That is the whole difference.

(3) From the following options (A–D), choose the one that most appropriately describes the interpretation of regression analysis results.

A If a prediction equation is obtained from regression analysis, the equation necessarily indicates a causal relationship.

B The results of regression analysis should be treated as one perspective that the data offers, and whether causation has been demonstrated must be carefully evaluated.

C If R^2 is high, the model is necessarily a good model.

D If the prediction equation fits, one may conclude that the explanatory variable is the cause of the response variable.

(4) From the following options (A–D), choose the one that is NOT an appropriate description of evaluation indicators for regression models.

A R^2 (coefficient of determination) takes values between 0 and 1; the closer to 1, the better the variation in the data is explained.

B RMSE is the square root of the mean of squared discrepancies between predicted and actual values, expressed in the same units as the response variable.

C MAE is the mean of the absolute values of the discrepancies between predicted and actual values, and is less affected by extremely large discrepancies.

D If the value of R^2 is close to 1, the model is necessarily a good model without needing to check other indicators.

PAUSE & THINK

A high R^2 alone — enough to trust a model, or do you need RMSE and MAE too?

Think as an Engineer — Investigate & Decide

Your school canteen wants to predict its daily cold-drink sales so it can order the right amount. You plan a regression model.

☐ **Trust one obvious cause** ☐ **Model it and test the model**

1 · Set up the model. Name the response variable, then list the explanatory variables you would try (for example the day's temperature, the number of students present, and whether it is an exam day). Decide whether this is a simple or a multiple regression, and say why.

2 · Judge and question it. Choose which evaluation indicator you would report (R^2, RMSE, or MAE) and justify it for this purpose. Then name one confounder (a confounding variable) — a hidden third factor that could make two variables move together without one causing the other — that could mislead you here.

3 · Decide and interpret. Suppose your model gives a high R^2. State whether that alone would let you claim "hot days cause higher sales," and what else you would need. **In pairs,** compare models and agree on the single most useful explanatory variable to keep.

Give evidence for your answer.

KEY TERMS

coefficient — the weight on an explanatory variable.
intercept — the constant *b* in the equation.
confounder — a hidden third cause behind a correlation.

Stuck? Start from the result you want to predict (sales), then ask which measurable things plausibly move it — and for each, ask "could something else be driving both?" A high R^2 tells you the line fits, not that the cause is real.

🌍 In a New Context

A city finds that months with more ice cream sales also have more swimming accidents, and a regression line fits the two closely. Explain why this does not mean ice cream causes accidents, and name the hidden factor most likely behind both.

❓ Lesson Question — Answered

The lesson asked how regression turns data into a prediction and how we judge whether it can be trusted. Regression analysis expresses the relationship between an explanatory variable (the input used to make the prediction) and a response variable (the result) as a model: a regression line $y = ax + b$ in simple regression (one explanatory variable), or a weighted sum such as $y = a_1x_1 + a_2x_2 + b$ in multiple regression (several variables), fitted by least squares to minimize the sum of squared residuals. Trust is judged with evaluation indicators used together — R^2 for how much variability is explained, and RMSE and MAE for the size of prediction errors (RMSE weighting large misses more, MAE less swayed by outliers) — never R^2 alone, and never by adding variables for their own sake. Above all, a good fit is only a correlation: causation requires temporal precedence, ruling out other factors, and theory. Regression offers one perspective from the data, to be interpreted, not obeyed.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

In regression analysis, the variable that is the result is called the (a), and the variable used to predict it is called the (b). The method that predicts using one

explanatory variable is called (c), and the method that predicts using multiple explanatory variables is called (d). To judge whether a model is good, evaluation indicators are used. Indicators include (e), which shows how well the model explains the variability in the data, and (f) and (g), which show the size of the discrepancies between predicted and actual values. To claim causation from the results of regression analysis, temporal precedence, exclusion of other factors' influence, theoretical support, and so on are necessary, and one must not conclude that there is (h) just because the prediction equation fits.

1. Fill in the blanks (a)–(h).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes correlation and causation.

A If there is correlation, causation necessarily holds.

B To claim causation, temporal precedence, exclusion of other factors' influence, theoretical support, and so on are necessary.

C If a prediction equation is obtained from regression analysis, the equation completely demonstrates causation.

D If there is no correlation, it can be said that there is no relationship at all between the two variables.

(2) From the following options (A–D), choose the one that most appropriately describes the interpretation of evaluation indicators.

A R^2 takes values between 0 and 1; the closer to 1, the better the variation in the data is explained.

B RMSE and MAE use the same calculation method, and their values always coincide.

C A model with a high R^2 is always a good model in any situation.

D MAE is an indicator that weights large discrepancies more heavily.

(3) From the following options (A–D), choose the one that is NOT an appropriate point of caution when using regression models.

A Do not conclude that there is causation just because the prediction equation fits.

B Confirm accuracy with evaluation indicators and verify the validity of the model in light of the purpose.

C If R^2 is close to 1, one may judge the goodness of the model without checking other indicators.

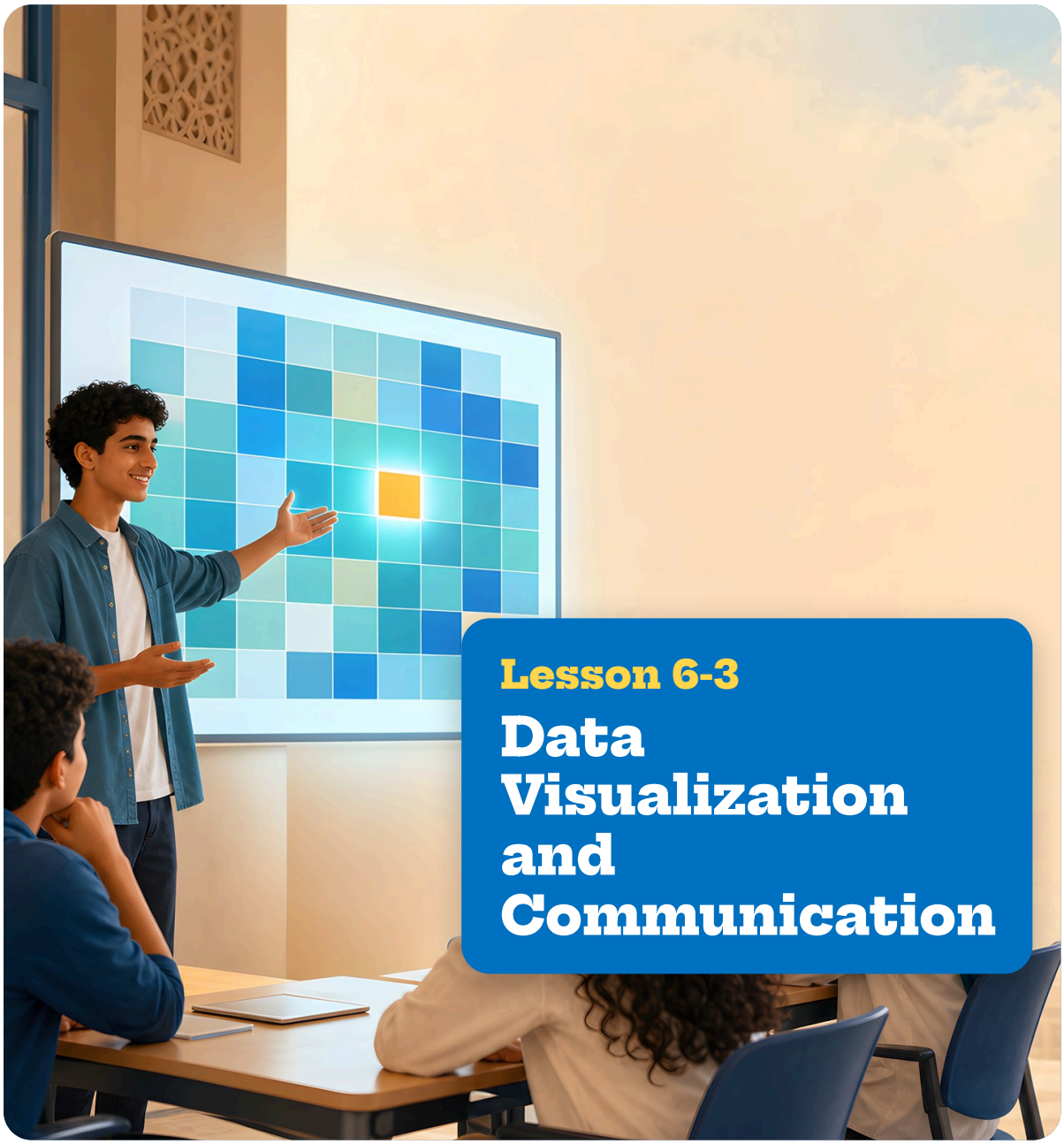
D Adding more explanatory variables does not necessarily improve accuracy. Appropriate variables must be chosen according to the purpose.

REFLECT & CHALLENGE

Reflect: Think of two things you have seen rise together (say, screen time and poor sleep). What hidden factor could drive both? **Challenge:** You may add only one more explanatory variable to a model that already has a high R^2. Argue whether you would add it at all, and state the one check you would run before trusting the bigger model.

⭐ Key takeaway

Remember:
Regression predicts a response variable from one (simple, $y = ax + b$) or several (multiple) explanatory variables, fitted by least-squares. Judge it with R^2, RMSE, and MAE together — and never read a good fit as proof of cause.



Lesson 6-3

Data Visualization and Communication

Learning Objectives

- Explain the role of data visualization and select suitable graphs, including for multiple variables.
- Apply data storytelling to communicate a clear and honest message from data.

A table of numbers can hide the very thing you most need to see. Turn it into the right picture and the pattern jumps out — the busiest hours, the strongest link, the odd outlier. Visualization does two jobs: it helps you understand your own data, and it helps you explain it to others. When one or two variables are involved, a bar, line, or scatter plot is enough; when three or more variables, or many combinations of information, must be shown at once, heatmaps and mosaic plots are used instead. And a chart alone is not a message: telling a clear data story is what makes an audience actually understand.

? Guiding question: How do we choose a visualization that reveals a pattern honestly, and how do we turn a chart into a message others understand?

Key Fact:
Visualization turns numbers into patterns the eye can grasp — to help the analyst think and to communicate. Common graphs suit one or two variables; heatmaps and mosaic plots handle more combinations. Good communication uses data storytelling: a clear message, only the elements that serve it, and honest labels.

Key Concepts:

- data visualization
- heatmap
- mosaic plot
- overplotting
- data storytelling

Point!

1. The Role of Data Visualization

(1) **(Data visualization)**…representing data as graphs or figures so that features that are hard to read from numbers alone can be grasped visually.

(2) Data visualization broadly has two roles.

Role	Description	Example
Deepen the analyst's own understanding	Visually grasp the distribution, trends, outliers, etc. of data and gain insight	When training records or study time are shown as a line graph, changes and trends that are hard to notice when numbers are merely listed in a table become easier to see
Communicate data to others	Convey the analysis results or claims to others in an easy-to-understand way	When the average test score of each class is shown as a bar graph, even someone seeing the data for the first time can compare differences between classes at a glance

(3) In either role, it is important to choose a graph that fits the content to be conveyed and to use representations that do not invite misunderstanding. The appropriate visualization method differs depending on the number and type of variables and the purpose of the analysis. When information that cannot be handled by familiar graphs needs to be shown, more advanced visualization techniques are used.

KEY TERMS
data visualization — showing data as graphs/figures to grasp features visually.



A clear chart speaks; a cluttered one hides.

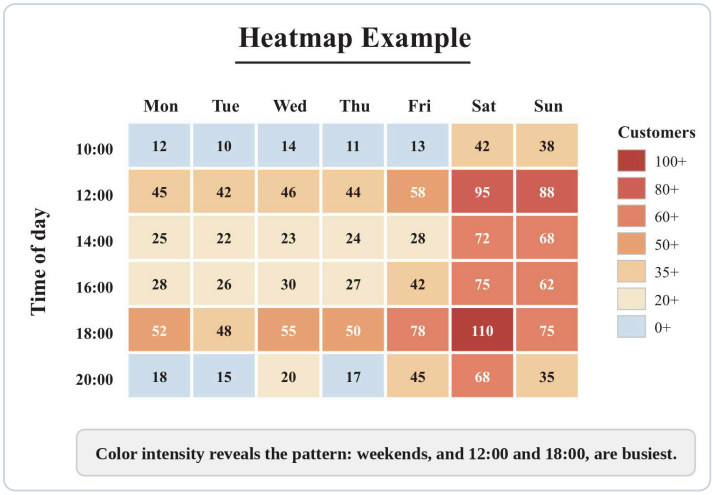
2. Visualization Handling Multiple Variables

Common graphs such as bar graphs, line graphs, and scatter plots are suited to representing the relationship between one or two variables. On the other hand, in analysis, situations often arise in which three or more variables, or many

KEY TERMS
heatmap — a color grid on two axes for a third value.
overplotting — points overlapping so density is hidden.

combinations of information, need to be handled at once. Heatmaps and mosaic plots are advanced visualization methods used for such cases.

(1) **(Heatmap)**…a graph in which items are arranged on two axes, and the size of a **(third value)** — a numerical value — corresponding to each combination is represented by the gradation of color or differences in hue. *e.g.,* Grasp the number of customers visiting at a glance by day of week × time slot; compare trends in average temperature for city × month.



A heatmap: color intensity shows the value for each row × column combination

In a scatter plot, since the relationship between two quantitative variables is represented by points, when the number of points becomes large, they overlap and the density of data becomes hard to see (overplotting). Also, scatter plots are not suited to handling combinations of categorical variables. In such cases, a heatmap has the advantage of allowing the density and patterns of the data to be grasped at a glance.

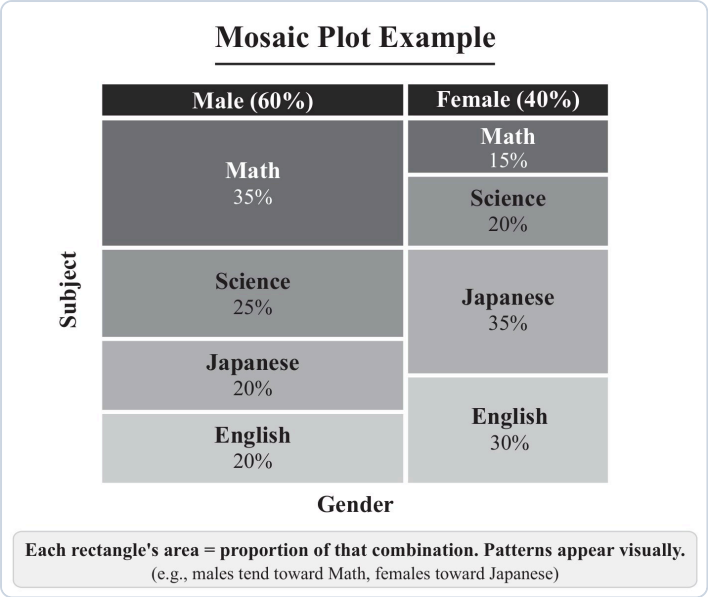
(2) **(Mosaic plot)**…a graph that represents the relationship between two categorical variables by the size of rectangular areas. *e.g.,* Compare the distribution of gender × subject preference; grasp differences in purchase ratios for age group × product category.



One view can encode several variables at once.

PAUSE & THINK

Two categories compared by area, or a third value by color — which graph is which?



A mosaic plot: each rectangle's area shows the proportion of that category combination

The widths of row and column divisions correspond to the proportions of frequencies, and the area of each rectangle represents the proportion of frequency for that combination of categories. Even when the overall picture is hard to grasp from a numerical table (cross-tabulation), a mosaic plot visually shows differences in proportions as areas, making it easier to grasp relationships between categories intuitively.

3. Data Storytelling

(1) **(Data storytelling)**…using data and visualization to convey the message you want to convey, structured like a story.

(2) The following are the three basic elements of storytelling.

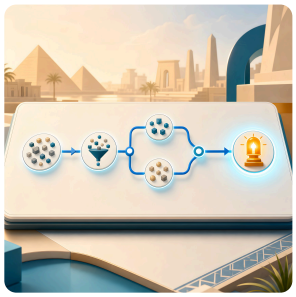
Element	Description	Example
Clarification of the message	Make it possible to express in one sentence what you want to convey through the graph	Make the title of a graph showing class test results contain the claim, such as “The class scored highly overall on the math test”
Selection of elements	Remove elements unrelated to the message and emphasize the parts you want to draw attention to	Make only the elements you want to draw attention to stand out in color, while keeping other elements in faded colors
Provision of supporting information	Properly arrange the title, axis labels, units, and notes	Make the graph title, such as “Average math test scores by class,” express the content at a glance, and clearly indicate units and item names like “points” on the vertical axis and “class” on the horizontal axis

(3) The depth of an audience's understanding can change greatly with how the same analysis result is conveyed. After visualizing data, it is important to put yourself in the audience's position and verify whether the message is conveyed.

KEY TERMS

mosaic plot — rectangle areas for two categorical variables.

data storytelling — structuring a data message like a story.



Storytelling guides the eye to one insight.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Data visualization has both the role of deepening the analyst's own understanding and the role of communicating data to others.

B A heatmap is a graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue.

C A mosaic plot is a graph that represents the relationship between two categorical variables by the size of rectangular areas.

D In data storytelling, it is important to include as many elements related to the graph as possible.

(2) Match each description (a–c) with the most appropriate graph from the choices below (A–C).

a A graph that represents the size of a numerical value corresponding to combinations on two axes by the gradation of color or differences in hue

b A graph that represents the relationship between two categorical variables by the size of rectangular areas

c A graph that represents the relationship between two quantitative variables by the position of points

[Choices] **A** Scatter plot **B** Heatmap **C** Mosaic plot

✓ Solution

(1)

A Correct as a role of data visualization. Therefore, ○.

B Correct definition of a heatmap. Therefore, ○.

C Correct definition of a mosaic plot. Therefore, ○.

D In data storytelling, it is important to remove elements unrelated to the message. Therefore, ×.

(2)

a: Showing values by gradation of color or hue is a heatmap. **B**

b: Showing relationships by area is a mosaic plot. **C**

c: Showing relationships by points is a scatter plot. **A**

PAUSE & THINK

Does adding more elements to a graph make the message clearer, or bury it?

Try

1 Answer the following questions.

1. What is the term for the graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue?
2. What is the term for the graph that represents the relationship between two categorical variables by the size of rectangular areas?
3. In data visualization, of “deepening the analyst's own understanding” and “communicating data to others,” for what purpose is the latter performed?
4. What is the term for using data and visualization to convey the message you want to convey, structured like a story?

2 Answer the following questions.

(1) For each of situations (a, b), choose the most appropriate graph from the choices below (A, B).

a You want to grasp the pattern of customer visits by day of week × time slot at a glance.

b You want to intuitively show differences in proportions for gender × subject preference using areas.

[Choices] **A** Heatmap **B** Mosaic plot

(2) From the following options (A–D), choose the one that most appropriately describes a situation in which a heatmap is used.

A A situation in which one wants to simply compare a representative value for one variable.

B A situation in which the relationship between two quantitative variables can be sufficiently grasped with a scatter plot because the number of points is small.

C A situation in which one wants to grasp the distribution and patterns of numerical data over combinations on two axes at a glance.

D A situation in which one wants to show only one composition ratio of a categorical variable.

EXAM-STYLE QUESTION

Explain why choosing the wrong graph, or hiding a chart's labels, can mislead an audience even when the data is correct. **[6] Refer to:** matching the graph to the data and honest supporting information.

PAUSE & THINK

Two axes with a value by color, or two categories by area? Name each situation's graph.

PAUSE & THINK

Time-series of one value, or proportions of two categories — which suits a mosaic plot?

(3) From the following options (A–D), choose the one that most appropriately describes a situation in which a mosaic plot is used.

A A situation in which one wants to follow the time-series changes of one quantitative variable.

B A situation in which one wants to intuitively grasp the proportions of frequencies for combinations of two categorical variables using areas.

C A situation in which one wants to represent the correlation between two quantitative variables by points.

D A situation in which one wants to show the composition ratio of categories at a particular time using sectors.

(4) From the following options (A–D), choose the one that most appropriately describes data storytelling.

A Even if the message you want to convey is unclear, it will be communicated to the audience as long as the graph is beautiful.

B Clarification of the message, selection of elements, and provision of supporting information are the basic elements.

C It is better to include as many related elements as possible in a graph.

D Titles and axis labels detract from the appearance of a graph and should preferably be omitted.

⚙ Think as an Engineer — Investigate & Decide

Your class collected data on students' home governorate and their favorite sport, and also monthly rainfall for several Egyptian cities across the year. You must present each dataset clearly.

☐ **Use a bar graph for everything**

☐ **Match each dataset to the right graph**

1 · Choose a graph for each. For the governorate × favorite-sport data and for the city × month rainfall data, name the graph you would use. For at least one of them, justify your choice — could the other graph type also work here, and why is yours better?

2 · Guard against misleading. Name one design choice that could make your chosen chart mislead the reader (for example a color scale with no legend, or truncated categories), and say how you would prevent it — a judgment the lesson's examples do not hand you.

3 · Tell the story. For one of the two charts, write the one-sentence message you want the audience to take away, and name the single element you would emphasize to carry it. **In pairs,** compare messages and agree on the clearer one.

Give evidence for your answer.

KEY TERMS

cross-tabulation — a table of counts for category combinations.

misleading graph — one whose design distorts the true pattern.

Stuck? Ask what each dataset is made of: two categories → areas (mosaic); a value across two axes → color grid (heatmap). Then ask what one sentence you want the reader to leave with, and strip away everything that does not serve it.

🌍 In a New Context

A researcher plots study time against test score for every student in a large school as a scatter plot, but so many points overlap into a solid blob that the density cannot be read. Name a visualization that would reveal where most students actually fall, and explain why it works better here than the crowded scatter plot.

❓ Lesson Question — Answered

The lesson asked how to choose a visualization that reveals a pattern honestly and how to turn a chart into a message others understand. Data visualization serves two roles — deepening the analyst's own understanding and communicating to others — and the right method depends on the number and type of variables. Bar, line, and scatter plots suit one or two variables; when a scatter plot's points overlap (overplotting) or categories must be combined, a heatmap shows a value across two axes by color, and a mosaic plot shows two categorical variables by rectangular area. Turning a chart into a message is data storytelling, built from three elements: clarify the message in one sentence, select only the elements that serve it (emphasizing the key part, fading the rest), and provide honest supporting information (title, axis labels, units, notes). Because the same result can land very differently depending on how it is shown, the final step is to stand in the audience's position and check that the message truly arrives.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Data visualization has both the role of deepening the analyst's own understanding and the role of (a) data to others. When information cannot be

handled by common graphs such as bar graphs and line graphs, advanced visualization techniques are used. The graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue is called (b), and the graph that represents the relationship between two categorical variables by the size of rectangular areas is called (c). When communicating data, the concept of (d), in which the message is structured like a story, is important; the basic elements are clarification of the message, selection of elements, and provision of (e).

1. Fill in the blanks (a)–(e).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes a heatmap.

A A graph that represents the relationship between two quantitative variables by the position of points.

B A graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue.

C A graph that represents the composition ratio of categories by the area of sectors.

D A graph that represents the changes in time-series data by lines.

(2) From the following options (A–D), choose the one that is NOT an appropriate basic element of data storytelling.

A Make it possible to express in one sentence what you want to convey through the graph (clarification of the message).

B Remove elements unrelated to the message and emphasize the parts you want to draw attention to (selection of elements).

C Properly arrange the title, axis labels, units, and notes (provision of supporting information).

D Include as many elements unrelated to the message as possible to increase the amount of information.

(3) For each of analysis purposes (a–c), choose the most appropriate graph from the choices below (A–C).

a You want to grasp the monthly average temperatures of multiple cities together at a glance.

b You want to show differences in purchase ratios for age group × product category using areas.

c You want to see the relationship between two quantitative variables in a case where the number of points is small enough that the density can also be sufficiently read.

[Choices] **A** Scatter plot **B** Heatmap **C** Mosaic plot

REFLECT & CHALLENGE

Reflect: Recall a chart that confused you. Was it the wrong graph, too many elements, or missing labels? **Challenge:** You are given a heatmap whose color scale does not start at zero, so small differences look dramatic. Describe the false impression this creates, and the one change that would make it honest without changing the data.

Key takeaway

Remember:
Match the graph to the data (heatmap for a value across two axes; mosaic plot for two categories by area; scatter/bar/line for one or two variables), then tell a data story: one message, only the elements that serve it, honest labels — and check it lands with the audience.